Simona Kocour

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Work Experience

Visiting PhD Student at ETH Zurich Aug 2025 - present

Working with Prof. Dr. Siyu Tang and Dr. Sergey Prokudin on Spatial Intelligence in 3D, semantic 3D scene understanding, and semantic segmentation.

PhD candidate in pan-European AI network of excellence Sept 2024 - present

Member of the ELLIS PhD program. Supervised by Dr. Torsten Sattler (CTU Prague) and Prof. Dr. Siyu Tang (ETH Zurich).

Researcher at Czech Technical University Apr 2024 - present

Member of Spatial Intelligence Group led by Dr. Torsten Sattler.
Research objective: Novel 3D scene representation with ability to control the level of detail.
Implementing 3D neural rendering and reconstruction (NeRF, Gaussian Splatting) Using 3D geometry and Structure from motion (SfM).

Machine Learning Researcher at Emplifi Nov 2020 - Mar 2024

Worked with large data sets from social media using PySpark, AWS, and Databricks.
Lead, and managed my projects, followed by presenting results to C-level.
Built pipeline and fine-tuned models for classification tasks (BERT, RoBERTa, DeepPavlov).
Built summarization and short text generation pipelines using GPT.
Behaviour scoring in time - built classification model with loss function emphasising customer behaviour with reflection on time.

Data Scientist at Aures Holdings Aug 2019 - Oct 2020

Solved business tasks using ML algorithms with Python and SQL.
Algorithmic pricing - built model for pricing goods automatically using general regression.
Behavioural scoring - built classification model.

Education

2024 - present	**PhD (Computer Vision)** at Czech Technical University
2017 - 2019	**Master's Degree in Mathematics & Statistics** at Masaryk University
2014 - 2017	**Bachelor's Degree in Mathematics** at Masaryk University

Publications

Kocour, S. et al. (2025). *Remove360: Benchmarking Residuals After Object Removal in 3D Gaussian Splatting*. arXiv: 2508.11431 [cs.CV]. URL: https://arxiv.org/abs/2508.11431.

Skills

Machine Learning:	3D Computer Vision, Neural Rendering, Neural Reconstruction, NLP PyTorch, TensorFlow
Software & Tools:	Python, R, Unix, Linux Docker, Git End-to-End Development, Software Prototyping
Data Processing:	AWS, Databricks, ElasticSearch, Presto PySpark, SQL
Languages:	Slovak (native) English, Czech (full professional proficiency) German (intermediate)